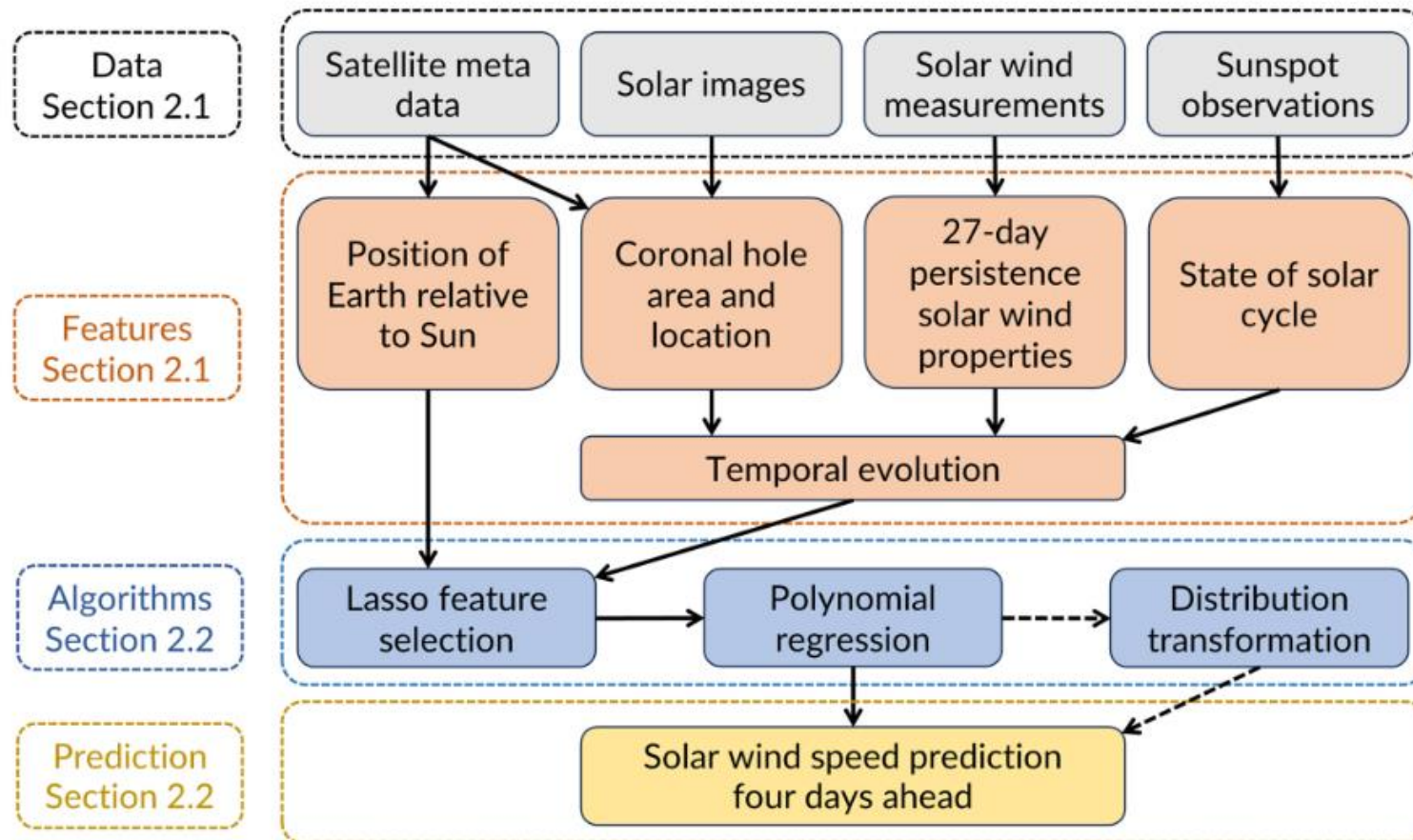
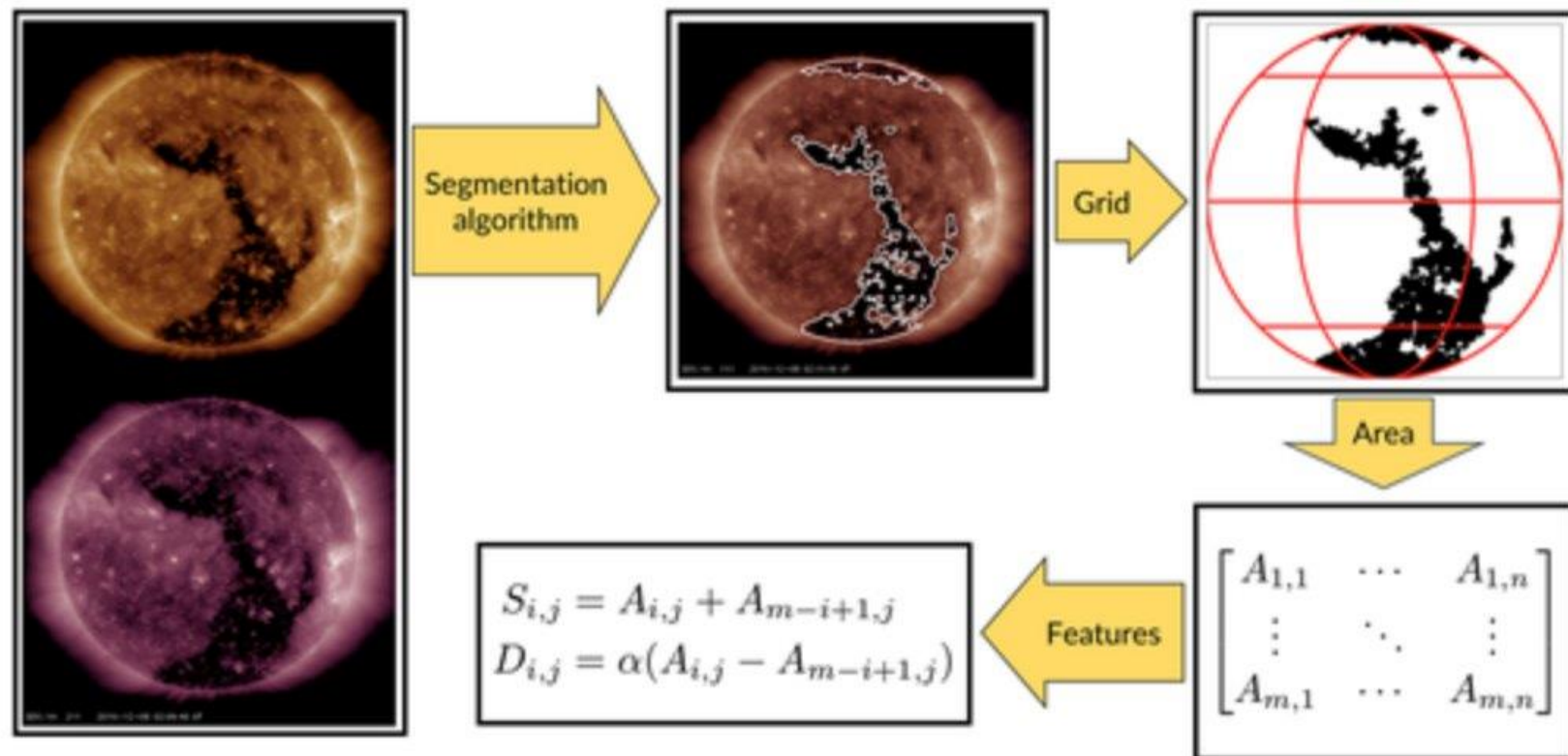


Collin, D., Shprits, Y., Hofmeister, S. J., Bianco, S., & Gallego, G. (2025). Forecasting high-speed solar wind streams from solar images. Space Weather





polynomial kernel of degree 3

Model	Timeline			HSS peak velocities			HSS events			
	RMSE	MAE	CC	RMSE	MAE	CC	POD	FAR	TS	BS
AV	97.4	76.2	-0.30	–	–	–	–	–	–	–
27	89.1	65.4	0.56	78.4	59.7	0.54	0.71	0.26	0.57	0.95
CH	71.2	54.8	0.67	123.4	102.8	0.52	0.69	0.11	0.64	0.78
Ours	68.1	52.6	0.70	87.5	66.8	0.62	0.77	0.13	0.69	0.88

Metrics for continuous time series:

		rmse	mae	me	cc
cv_set	eval_data_mode				
test	no_cme	68.418619	52.584346	1.482914e+00	0.694746
	trans	74.843774	57.498248	2.535792e+00	0.695511
	with_cme	75.483646	57.868315	-5.032541e+00	0.617334
	trans_with_cme	82.253344	63.167960	-4.408194e+00	0.615354
train	no_cme	64.372272	49.424132	-1.136868e-13	0.715008
	trans	69.923853	53.364642	1.367201e-01	0.711809
	with_cme	73.923729	56.260100	-6.642358e+00	0.617637
	trans_with_cme	80.180030	61.074445	-5.906838e+00	0.613365

#####

Prediction efficiencies for continuous time series:

		pe_av	pe_27	pe_base
cv_set	eval_data_mode			
test	no_cme	0.479520	0.409879	0.077541
	trans	0.377174	0.293838	-0.103849
	with_cme	0.371635	0.406760	0.056827
	trans_with_cme	0.253872	0.295579	-0.119935
train	no_cme	0.511201	0.462675	0.100805
	trans	0.423255	0.365999	-0.060979
	with_cme	0.374357	0.435268	0.066811
	trans_with_cme	0.263977	0.335634	-0.097828

#####

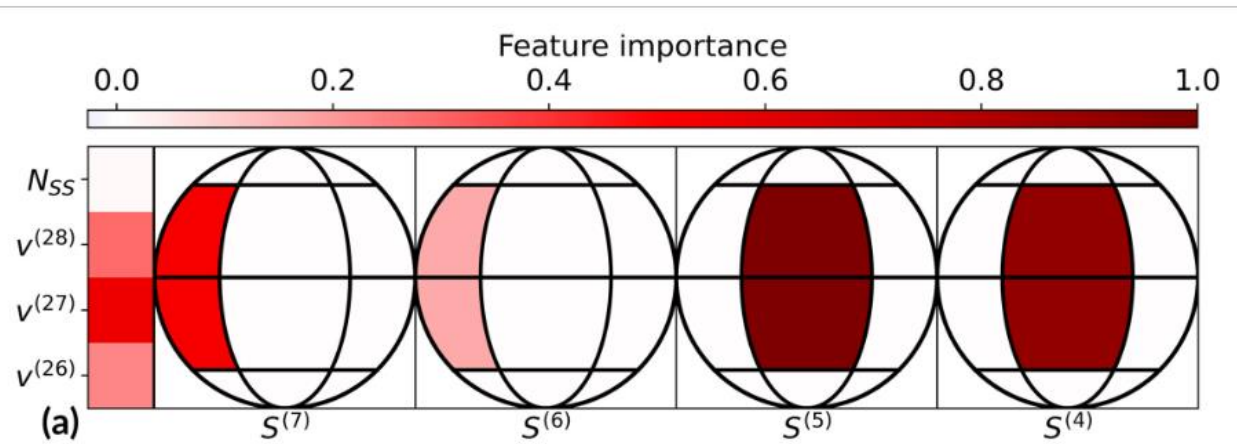
Metrics for associated HSS peaks:

		rmse	mae	me	cc
cv_set	eval_data_mode				
test	no_cme	113.739847	94.973590	-86.241787	0.570370
	trans	87.269625	67.450196	-54.207781	0.618002

#####

Metrics for HSS detection:

		pod	far	ts	bs
cv_set	eval_data_mode				
test	no_cme	0.721088	0.094017	0.670886	0.795918
	trans	0.768707	0.110236	0.701863	0.863946



Training time grows exponentially when the number of features increases.

	fold_1	fold_2	fold_3	fold_4	fold_5	cv_average
S[2,2](-4)	6.823836	6.185626	0.534262	0.000000	16.282822	5.965309
S[2,2](-5)	6.815916	2.913594	-0.500414	15.372513	4.307810	5.781884
speed(-27)	0.998666	4.769717	1.838614	6.574128	1.663619	3.168949
speed(-28)	0.201842	0.606901	1.111473	4.547166	2.433571	1.780191
speed(-26)	0.267573	0.000000	0.722973	3.610880	2.578655	1.436016
S[2,1](-7)	0.000000	0.000000	0.000000	6.464659	0.000000	1.292932
sunspot_number	0.000000	0.000000	0.000000	0.000000	-0.005144	-0.001029

--- permutation_importance_peak ---						
	fold_1	fold_2	fold_3	fold_4	fold_5	cv_average
S[2,2](-4)	49.429266	41.727585	34.052484	0.000000	60.829347	37.207736
S[2,2](-5)	23.954829	19.114869	17.846049	56.865046	23.947136	28.345586
speed(-27)	26.930107	31.423862	14.073176	37.801000	17.298007	25.505230
speed(-26)	17.056939	0.000000	14.412099	21.546216	16.801512	13.963353
S[2,1](-7)	0.000000	0.000000	0.000000	36.118904	0.000000	7.223781
speed(-28)	1.701100	4.359643	6.823243	2.941557	11.329377	5.430984
sunspot_number	0.000000	0.000000	0.000000	0.000000	0.376863	0.075373

Excluding past SWS from input

Metrics for continuous time series:

		rmse	mae	me	cc
cv_set	eval_data_mode				
test	no_cme	68.418619	52.584346	1.482914e+00	0.694746
	trans	74.843774	57.498248	2.535792e+00	0.695511
	with_cme	75.483646	57.868315	-5.032541e+00	0.617334
	trans_with_cme	82.253344	63.167960	-4.408194e+00	0.615354
train	no_cme	64.372272	49.424132	-1.136868e-13	0.715008
	trans	69.923853	53.364642	1.367201e-01	0.711809
	with_cme	73.923729	56.260100	-6.642358e+00	0.617637
	trans_with_cme	80.180030	61.074445	-5.906838e+00	0.613365

#####

Prediction efficiencies for continuous time series:

		pe_av	pe_27	pe_base
cv_set	eval_data_mode			
test	no_cme	0.479520	0.409879	0.077541
	trans	0.377174	0.293838	-0.103849
	with_cme	0.371635	0.406760	0.056827
	trans_with_cme	0.253872	0.295579	-0.119935
train	no_cme	0.511201	0.462675	0.100805
	trans	0.423255	0.365999	-0.060979
	with_cme	0.374357	0.435268	0.066811
	trans_with_cme	0.263977	0.335634	-0.097828

#####

Metrics for associated HSS peaks:

		rmse	mae	me	cc
cv_set	eval_data_mode				
test	no_cme	113.739847	94.973590	-86.241787	0.570370
	trans	87.269625	67.450196	-54.207781	0.618002

#####

Metrics for HSS detection:

		pod	far	ts	bs
cv_set	eval_data_mode				
test	no_cme	0.721088	0.094017	0.670886	0.795918
	trans	0.768707	0.110236	0.701863	0.863946

Metrics for continuous time series:

		rmse	mae	me	cc
cv_set	eval_data_mode				
test	no_cme	71.023923	54.808694	-0.230026	0.663283
	trans	79.638603	61.233841	0.920433	0.655965
	with_cme	77.940728	59.455485	-10.131014	0.586755
	trans_with_cme	86.349648	66.203660	-11.580728	0.580775
train	no_cme	66.636662	51.581910	0.000000	0.690104
	trans	73.299494	56.762813	0.214242	0.683647
	with_cme	76.161951	57.792139	-11.444425	0.593302
	trans_with_cme	83.266124	63.678703	-13.280181	0.588858

#####

Prediction efficiencies for continuous time series:

		pe_av	pe_27	pe_base
cv_set	eval_data_mode			
test	no_cme	0.439127	0.972058	0.097786
	trans	0.294816	0.964869	-0.134352
	with_cme	0.330061	0.967113	0.074863
	trans_with_cme	0.177705	0.959634	-0.135530
train	no_cme	0.476207	0.975152	0.135496
	trans	0.366225	0.969935	-0.046026
	with_cme	0.335898	0.968471	0.085971
	trans_with_cme	0.206229	0.962315	-0.092498

#####

Metrics for associated HSS peaks:

		rmse	mae	me	cc
cv_set	eval_data_mode				
test	no_cme	125.003035	105.889789	-103.042851	0.537528
	trans	94.565453	74.520374	-61.679122	0.549346

#####

Metrics for HSS detection:

		pod	far	ts	bs
cv_set	eval_data_mode				
test	no_cme	0.727891	0.144000	0.648485	0.850340
	trans	0.782313	0.178571	0.668605	0.952381

Replacing polynomial kernel with MLP

```
options_mlp_conservative = {  
    'prediction_model': 'mlp',  
    'mlp_hidden_layers': (64, 32),  
    'mlp_learning_rate': 0.0003,  
    'mlp_max_iter': 2000,  
    'mlp_batch_size': 128,  
    'mlp_use_cuda': True,  
    'mlp_weight_decay': 0.1,  
    'mlp_dropout_rate': 0.6,  
}
```

Metrics for continuous time series:

		rmse	mae	me	cc
cv_set	eval_data_mode				
test	no_cme	70.714848	53.624014	0.047247	0.669727
	trans	77.779039	58.866105	-2.297951	0.670882
	with_cme	77.097651	58.652400	-5.945013	0.597068

Metrics for associated HSS peaks:

		rmse	mae	me	cc
cv_set	eval_data_mode				
test	no_cme	108.219114	90.444846	-82.837968	0.559435
	trans	81.800471	64.914269	-49.821888	0.588317

```
options_mlp_regularized = {  
    'prediction_model': 'mlp',  
    'mlp_hidden_layers': (256, 128, 64),  
    'mlp_learning_rate': 0.0005,  
    'mlp_max_iter': 2000,  
    'mlp_batch_size': 256,  
    'mlp_use_cuda': True,  
    'mlp_weight_decay': 0.01, # L2 regul  
    'mlp_dropout_rate': 0.4, # 40% dr  
}
```

Results of full cross-validation averaged over all 5 folds:

#####

Metrics for continuous time series:

		rmse	mae	me	cc
cv_set	eval_data_mode				
test	no_cme	74.304595	55.432145	-2.512506	0.636473
	trans	79.802956	59.517287	-0.768211	0.638988
	with_cme	80.946666	60.746550	-8.114600	0.561383
	trans_with_cme	87.207862	65.609320	-6.023479	0.561050

		rmse	mae	me	cc
cv_set	eval_data_mode				
test	no_cme	86.448575	69.712450	-37.877280	0.469950
	trans	75.261889	60.435711	-21.956655	0.501608

- When using the same (less than 10) number of features, it is not surprising that a shallow MLP behaves similar to the polynomial kernel.
- Left to check: whether using wider and deeper MLP would improve the prediction with all other parameter fixed.
- If more features are used, the computational cost for polynomial kernel grows exponentially so a MLP would be a better option since it can approximate more complex functions.

What other features to include

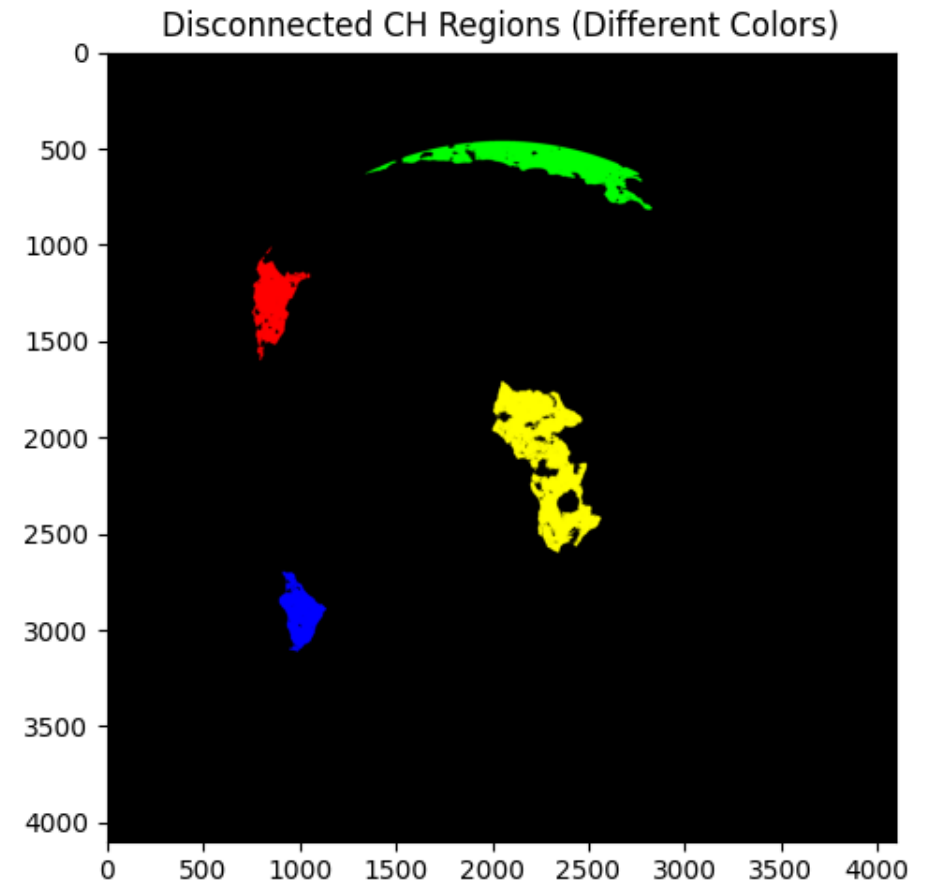
- Feature extractors on solar images
- Information related with intensities of solar spectrum./contrast

The Spatial Possibilistic Clustering Algorithm (SPoCA)

This gives us information of the locations (in terms of pixel coordinates) of CH and non-CH areas.

We expect that other than area, energy intensity of CH will also impact solar wind and/or high-speed streams.

The files are in .fits format. Does it make sense to download all files to covert and store locally?



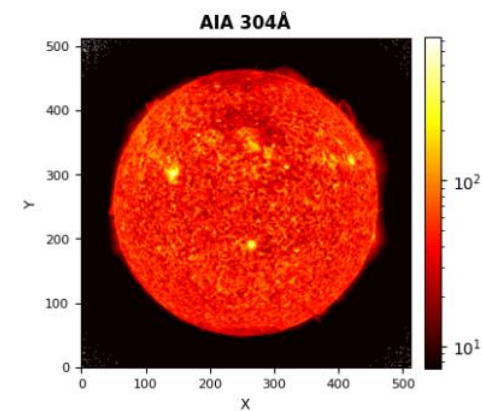
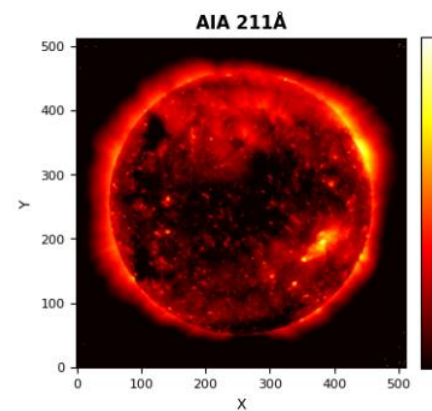
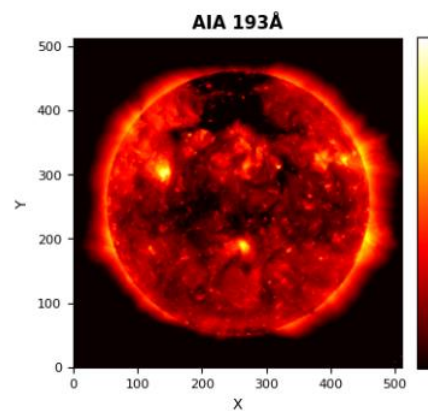
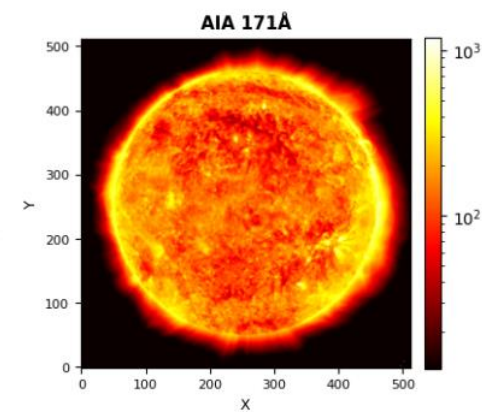
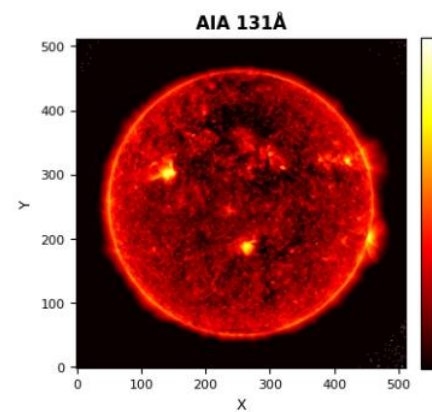
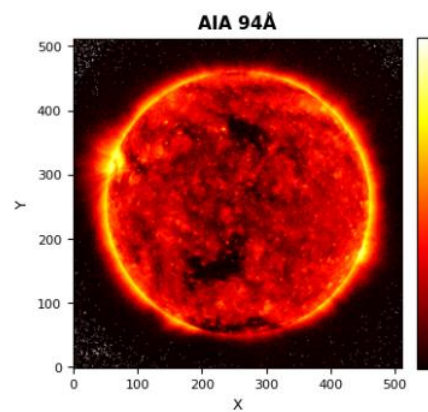
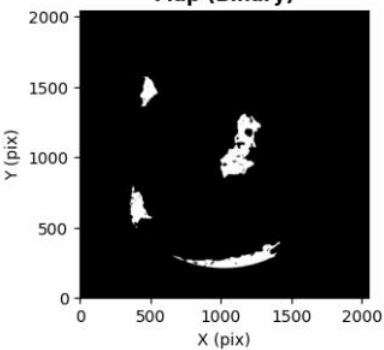
Page 1: Coronal Hole Map vs AIA Channels (131-211)
CH Date: 2010-05-13 | AIA Date: 2010-05-13 | Index: 8934

CORONAL HOLE MAP

Shape: (2048, 2048)
Date: 2010-05-13
RSUN: 949.8"

Statistics:
Min: 0.0000
Max: 1.0000
Mean: 0.0301
CH pix: 129,318

Coronal Hole
Map (Binary)



How would intensity play a role as input features

- As shown in the feature importance map before, CH at certain location and time delta are less relevant. One reasonable strategy can be to ignore the top and bottom edges.
- Total/average intensity of one/all CHs.
- Instant/average intensity over a time window
- Contrast of intensity of CH and non-CH